Lecture Unit 10.1

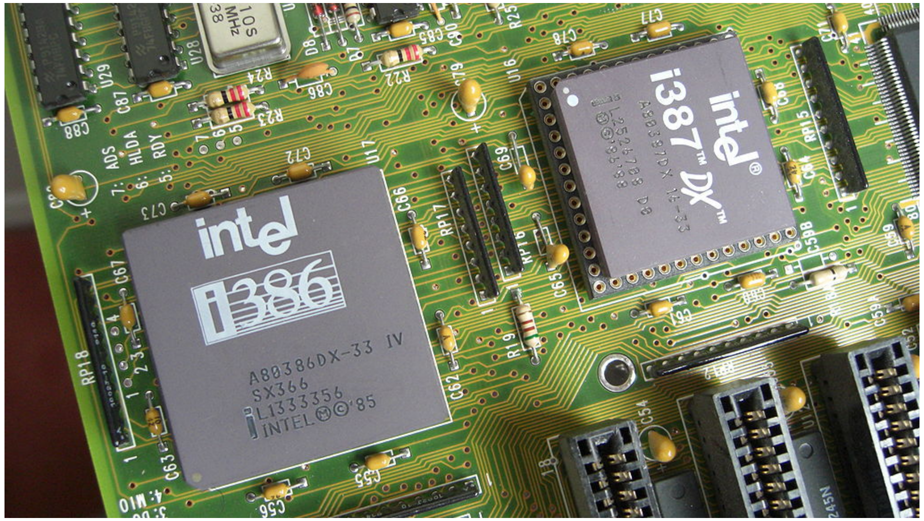
Background and history of accelerators

CPUs and accelerators

- CPUs need to deliver acceptable performance for a very wide range of applications
- CPU design is a trade off between functionality, performance, energy efficiency and cost
- Accelerators and co-processors work alongside a CPU to provide increased performance for specific workloads - different design trade offs

Accelerator development

- Using an accelerator to supplement the CPU is not a new idea
- Early x86 CPUs did not have floating point hardware
- x87 floating point co-processor was used to accelerate floating point operations
- Floating point units are now part of the main processor



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GPUs

- GPUs (Graphics Processing Units) are specialised hardware for carrying out operations relating to image generation
- Project 3D objects to form images → many matrix-vector operations
- This is a highly parallel task:
  - GPUs contain a large number of floating point units
  - Support a large number of processing threads
- Higher memory bandwidth than a CPU (different type of memory)

GPU development

- Early GPUs had a fixed function rendering pipeline
- Hardware evolved to support programmable “unified shaders” and double precision arithmetic
  - AMD RV670 (2007)
  - NVIDIA GT200 (2008)
- Current GPUs can be used for general purpose computation

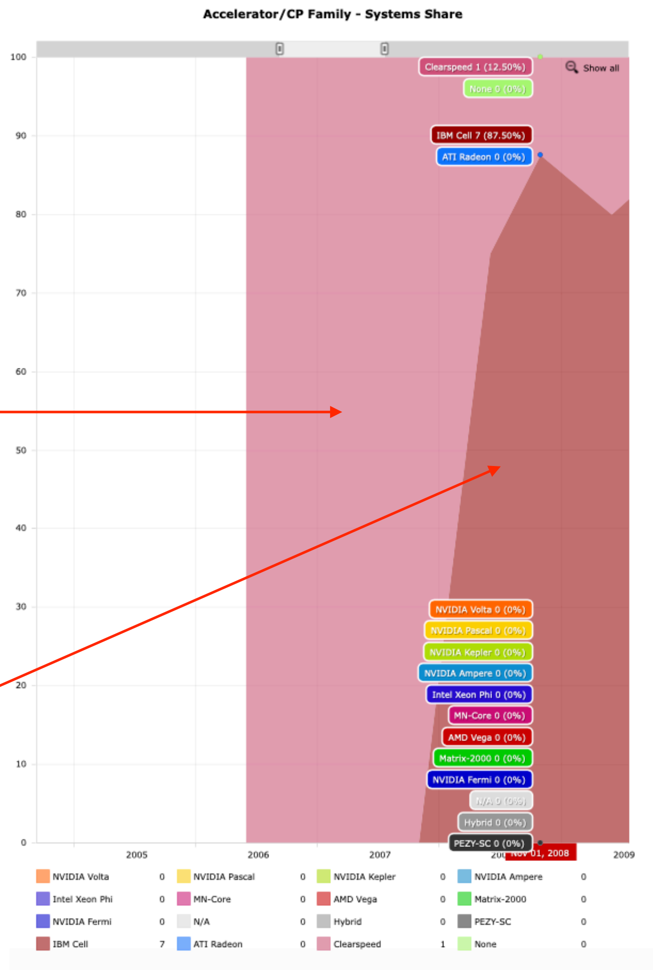
GPUs as accelerators

- Good floating point performance and high memory bandwidth are desirable characteristics for HPC applications
- GPU works as an accelerator for a host CPU
- Some operations need to be handled by the CPU e.g. running operating system, I/O
- Computationally expensive work can be offloaded to the GPU

Workshop 1: Accelerators and Co-Processors

Accelerators Family

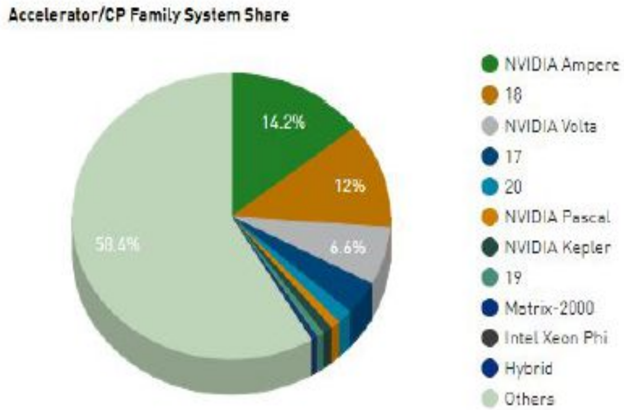
- What was the first accelerator to appear on the list and when did it appear?
    - Clearspeed (June 2006): floating point accelerator card
- What was the second accelerator to appear on the list?
    - IBM Cell (June 2008): as used in the Playstation 3



Accelerator family

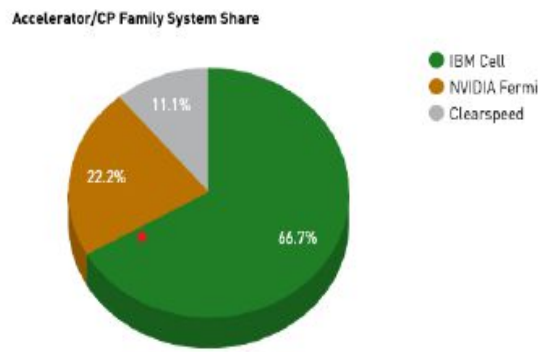
- What type of accelerators are most popular now? When did they start to appear?

November 2024



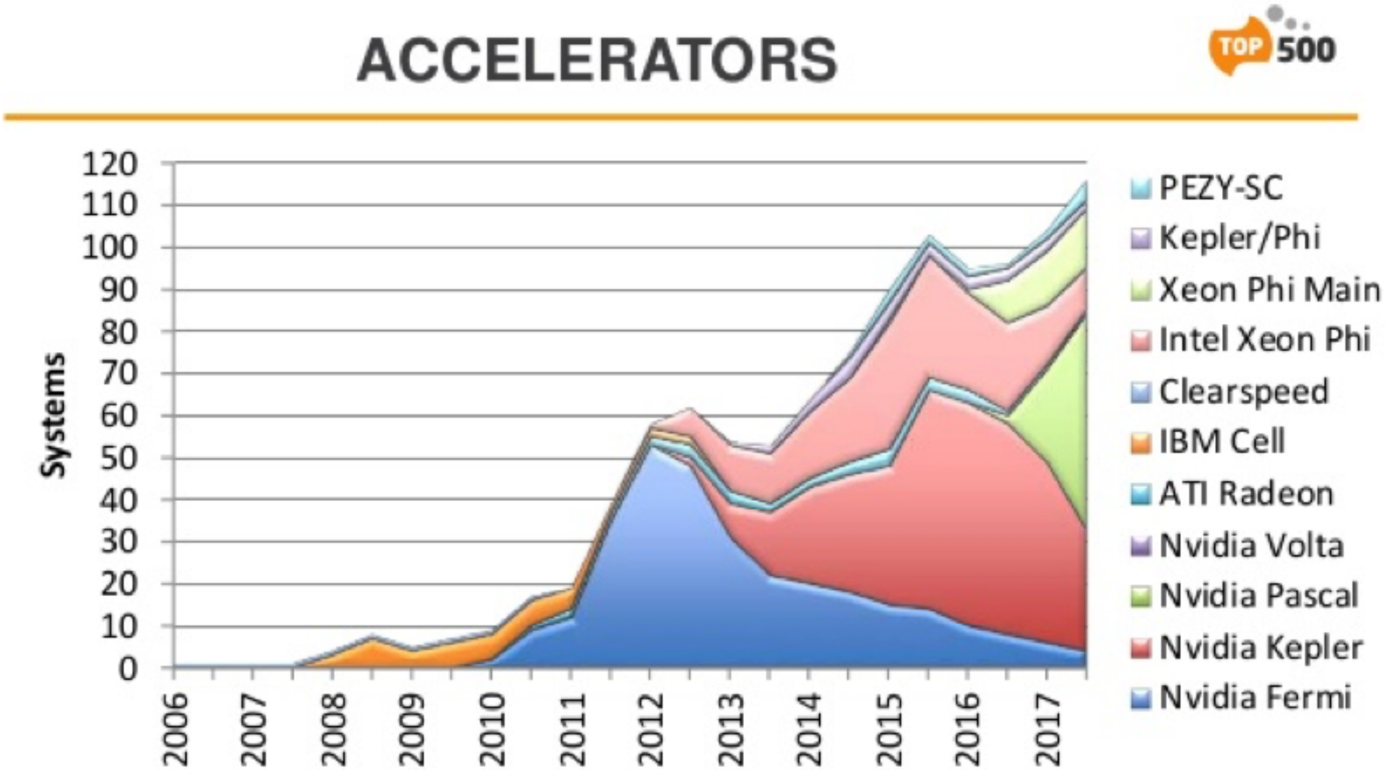
Mostly NVIDIA GPUs today

June 2010



These first appeared in the June 2010 list

Impact on top 500 list



What fraction of systems on the current list make use of accelerators?

	Accelerator/CP Family	Count	System Share (%)	Rmax (GFlops)	Rpeak (GFlops)	Cores
1	Others	292	58.4	2,036,993,986	0	0
2	NVIDIA Ampere	71	14.2	1,048,725,140	1,448,388,930	9,746,476
3	18	60	12	1,967,575,660	2,965,058,760	7,080,568
4	NVIDIA Volta	33	6.6	318,290,610	484,815,792	5,894,200
5	17	19	3.8	4,444,901,070	6,632,630,582	29,086,784
6	20	8	1.6	721,794,000	958,832,260	3,574,896
7	NVIDIA Pascal	5	1	42,902,620	60,652,584	839,200
8	NVIDIA Kepler	5	1	15,412,000	25,614,070	347,984
9	19	4	0.8	1,054,038,950	2,074,398,570	9,613,760
10	Matrix-2000	1	0.2	61,444,500	100,678,664	4,981,760
11	Intel Xeon Phi	1	0.2	2,539,130	3,388,032	194,616
12	Hybrid	1	0.2	3,126,240	5,610,481	152,692

Unit summary

- Accelerators can be used in combination with a CPU to speed-up computationally expensive tasks
- GPUs have good floating point performance and memory bandwidth
- GPUs are often used as accelerators in current HPC systems
- See SAB sections 15.1-15.4